Computing the SVD by Hand

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Where this fits: Pipeline map step 0 (Foundations). See the Course map.

0 Foundations	1 Frame	2 Get data	3 Understand	4 Clean	5 Features	6 Reduce
7 Split	8 Model	9 Evaluate	10 Tune	11 Deploy	12 Test	13 Monitor

- **Builds on:** Singular value decomposition.
- **Compare with:** Eigenvectors and eigenvalues.

1. Overview

Key point: A matrix stretches some directions more than others. To find the SVD by hand, we multiply the matrix by its own transpose. That new matrix keeps the stretching but forgets the final turn, so its eigenvectors are the directions of most and least stretch, and its **eigenvalues** (G-665) are the squared stretches.

The rotate, stretch, rotate picture showed what $A = U\Sigma V^\mathsf{T}$ means: turn, stretch, turn. The small T marks the **transpose** (G-2012): the same numbers with rows and columns swapped, so row 1 becomes column 1. For example:

$$\begin{bmatrix} 3 & 0 \\ 4 & 5 \end{bmatrix}^\mathsf{T} = \begin{bmatrix} 3 & 4 \\ 0 & 5 \end{bmatrix}$$

The three factors are:

- V, whose columns $\mathbf{v}_1, \mathbf{v}_2$ are the input directions (the **right singular vectors**, G-1692);
- Σ, the **diagonal matrix** (G-601) of stretches σ_1, σ_2 (the **singular values**, G-1812);
- U, whose columns $\mathbf{u}_1, \mathbf{u}_2$ are the output directions (the **left singular vectors**, G-1079).

It gave the factors of

$$A = \begin{bmatrix} 3 & 0 \\ 4 & 5 \end{bmatrix}$$

without saying where they came from. This Note finds them, with a four-step recipe:

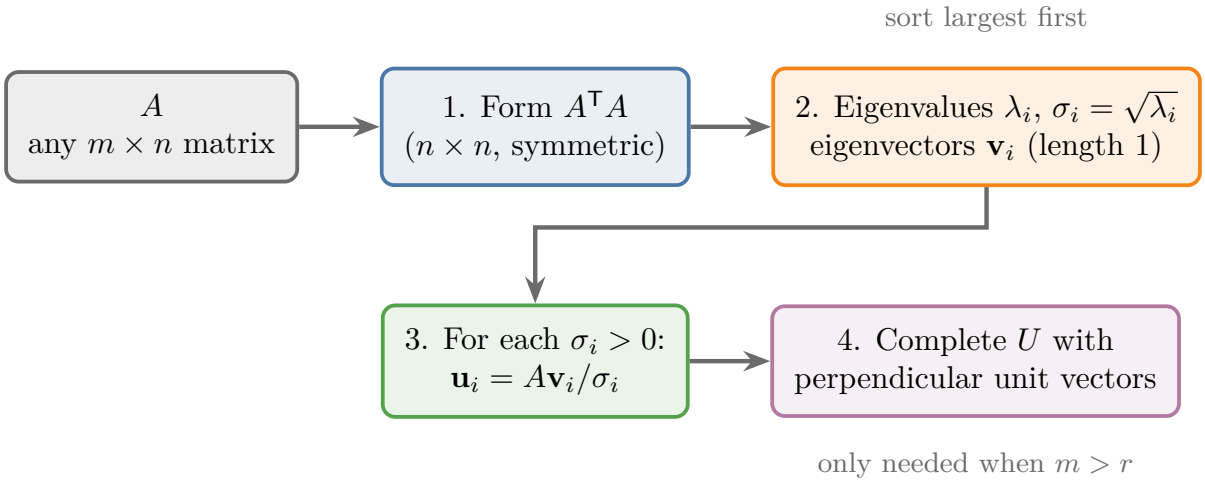


Figure 1: The four steps of computing an SVD by hand

Figure 1 is the plan for the whole Note. Read it top to bottom: form a new **symmetric matrix** (G-1932), take its eigenvectors and eigenvalues, turn the eigenvalues into stretches, then get the output directions from the input directions.

This Note covers:

- why $A^\mathsf{T}A$ and AA^T hold the answer (Section 2);
- the four-step recipe of Figure 1, worked on A (Section 3);
- a sign trap, and how computing each $\mathbf{u}_i$ from its $\mathbf{v}_i$ avoids it (Section 4);
- a matrix of **rank** (G-1627) 1, where a singular value is 0 (Section 5);
- the four subspaces that the SVD gives bases for (Section 6);
- a matrix that is not square (Section 7);
- why computers do not use this recipe (Section 8).

Finding eigenvalues and **eigenvectors** (G-666) by hand, with the determinant, is taught in finding eigenvalues; we use it here without repeating it.

2. Making one orthogonal matrix disappear

> **Key point:** The SVD has two unknown turning matrices, U and V. In $A^\mathsf{T}A$, the turn U cancels out, leaving a **symmetric matrix** (G-1932) whose eigenvectors and eigenvalues we can find by hand.

2.1 $A^\mathsf{T}A$ gives V and the singular values

> **Key point:** The directions in which A stretches most and least are the eigenvectors of $A^\mathsf{T}A$; the squared stretches are its eigenvalues.

The idea in plain words. Feed A every arrow of length 1 and measure how long each output is. Some input directions get stretched a lot, others only a little. The longest and shortest stretches, and the directions that get them, are exactly the singular values and the right singular vectors.

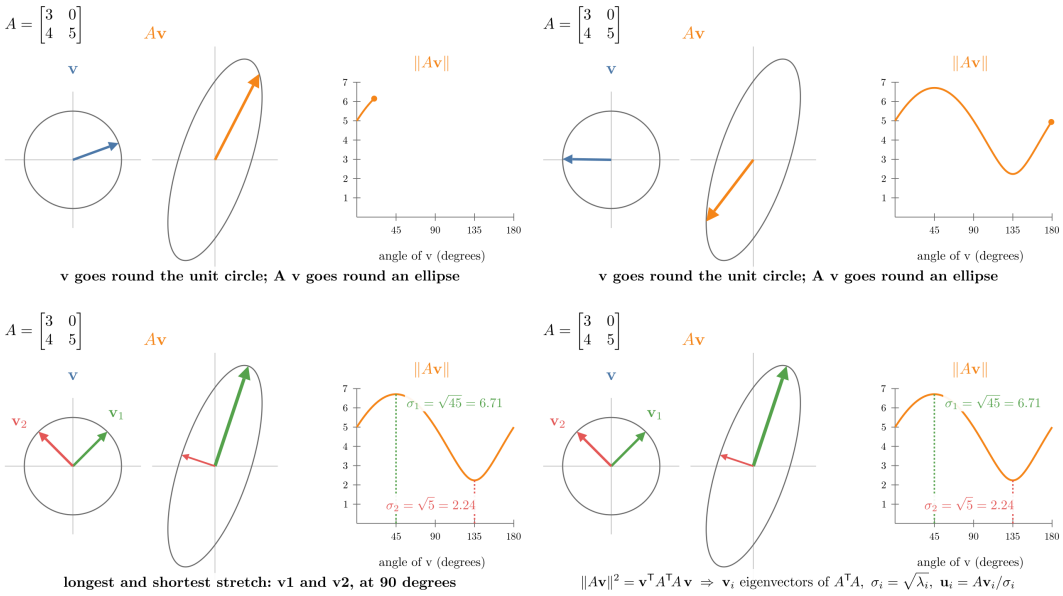


Figure 2: A unit vector $\mathbf{v}$ sweeps the circle and $A\mathbf{v}$ sweeps an ellipse; the length of $A\mathbf{v}$ peaks at $\sigma_1 = \sqrt{45}$ along $\mathbf{v}_1$ and bottoms out at $\sigma_2 = \sqrt{5}$ along $\mathbf{v}_2$, at 90° to $\mathbf{v}_1$

Figure 2 does this for A. A unit arrow $\mathbf{v}$ goes round the circle (left), its output $A\mathbf{v}$ goes round an ellipse (middle), and the graph (right) tracks the length of $A\mathbf{v}$. Watch the peak and the trough:

- the longest stretch, 6.71, comes at the direction $\mathbf{v}_1$;
- the shortest, 2.24, comes at $\mathbf{v}_2$, at a right angle to $\mathbf{v}_1$.

The two directions, each scaled to length 1:

$$\mathbf{v}_1 = [1, 1]/\sqrt{2}$$

$$\mathbf{v}_2 = [-1, 1]/\sqrt{2}$$

The question is how to find these two directions without sweeping every arrow.

The trick on numbers. Write $\|A\mathbf{v}\|$ for the length of the output $A\mathbf{v}$. For a column $\mathbf{x}$, the product $\mathbf{x}^\mathsf{T}\mathbf{x}$ is the dot product of $\mathbf{x}$ with itself, the squared length. For $\mathbf{x} = [3, 4]$:

$$\mathbf{x}^\mathsf{T}\mathbf{x} = 3 \times 3 + 4 \times 4$$

$$= 25$$

so its length is 5, the square root of 25. The squared length of an output can therefore be regrouped, using the rule $(A\mathbf{v})^\mathsf{T} = \mathbf{v}^\mathsf{T}A^\mathsf{T}$ (the **transpose** (G-2012) of a product is the product of the transposes in reverse order; Strang §2.7):

$$\|A\mathbf{v}\|^2 = (A\mathbf{v})^\mathsf{T}(A\mathbf{v})$$

$$= \mathbf{v}^\mathsf{T}A^\mathsf{T}A\mathbf{v}$$

$$= \mathbf{v}^\mathsf{T}(A^\mathsf{T}A)\mathbf{v}$$

So the matrix $A^\mathsf{T}A$ measures squared stretch. Here A^T is the transpose of A, worked out in Section 1. Compute $A^\mathsf{T}A$ entry by entry:

$$A^\mathsf{T} = \begin{bmatrix} 3 & 4 \\ 0 & 5 \end{bmatrix}$$

Entry of $A^\mathsf{T}A$	Row of A^T times column of A	Value
row 1, column 1	$3 \times 3 + 4 \times 4$	25
row 1, column 2	$3 \times 0 + 4 \times 5$	20
row 2, column 1	$0 \times 3 + 5 \times 4$	20
row 2, column 2	$0 \times 0 + 5 \times 5$	25

$$A^\mathsf{T}A = \begin{bmatrix} 25 & 20 \\ 20 & 25 \end{bmatrix}$$

Now apply $A^\mathsf{T}A$ to the peak direction $\mathbf{v}_1$ of Figure 2 (the vector $[1, 1]$, scaled to length 1):

$$A^\mathsf{T}A \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 25 + 20 \\ 20 + 25 \end{bmatrix}$$

$$= \begin{bmatrix} 45 \\ 45 \end{bmatrix}$$

$$= 45 \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

The direction did not change; it was only scaled by 45. So $\mathbf{v}_1$ is an **eigenvector** (G-666) of $A^\mathsf{T}A$ with **eigenvalue** (G-665) 45. And

$$\sqrt{45} = 6.71$$

is the peak stretch of Figure 2. The same check on the trough direction $[-1, 1]$:

$$A^\mathsf{T}A \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} -25 + 20 \\ -20 + 25 \end{bmatrix}$$

$$= \begin{bmatrix} -5 \\ 5 \end{bmatrix}$$

$$= 5 \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

$$\sqrt{5} = 2.24$$

the trough stretch.

Why it works: the algebra, one step per line. Start from $A = U\Sigma V^\mathsf{T}$. The transpose of a product is the product of the transposes in reverse order (as in Section 2.1), and $U^\mathsf{T}U = I$, the **identity matrix** (G-915), because U is an **orthogonal matrix** (G-1407) (see what an orthogonal matrix does to space):

$$A^\mathsf{T} = (U\Sigma V^\mathsf{T})^\mathsf{T} = V\Sigma^\mathsf{T}U^\mathsf{T}$$

$$A^\mathsf{T}A = V\Sigma^\mathsf{T}U^\mathsf{T}\,U\Sigma V^\mathsf{T}$$

$$= V\Sigma^{\mathsf{T}}\, I\, \Sigma V^{\mathsf{T}}$$

$$= V \begin{bmatrix} \sigma_1^2 & & \\ & \ddots & \\ & & \sigma_n^2 \end{bmatrix} V^{\mathsf{T}}$$

The right side is exactly an eigen-decomposition PDP^{-1} (see changing to an eigenbasis), with $P = V$ and $P^{-1} = V^{\mathsf{T}}$ (Strang §7.2; MML §4.5.2). So:

- the right singular vectors $\mathbf{v}_i$ are the eigenvectors of $A^{\mathsf{T}}A$;
- the squared singular values σ_i^2 are its eigenvalues λ_i, so $\sigma_i = \sqrt{\lambda_i}$.

Check: for A, the eigenvalues 45 and 5 give $\sigma_1 = 6.71$ and $\sigma_2 = 2.24$, the peak and trough of Figure 2.

2.2 Why this always works

> **Key point:** $A^{\mathsf{T}}A$ is symmetric with **eigenvalues** (G-665) of at least 0, so it always has perpendicular eigenvectors and real square roots.

A itself may have no useful **eigenvectors** (G-666), but $A^{\mathsf{T}}A$ always does, for two reasons.

It is symmetric. The matrix $A^{\mathsf{T}}A$ of Section 2.1 reads the same across its diagonal: 20 above and 20 below. That always happens, because its **transpose** (G-2012) is itself:

$$(A^{\mathsf{T}}A)^{\mathsf{T}} = A^{\mathsf{T}}(A^{\mathsf{T}})^{\mathsf{T}} = A^{\mathsf{T}}A$$

A **symmetric matrix** (G-1932) always has a full set of perpendicular eigenvectors (the same fact that makes PCA work on a covariance matrix, see the eigenvectors of the covariance matrix). So V can always be built as an **orthogonal matrix** (G-1407). Check: the dot product of $[1, 1]$ and $[-1, 1]$ is

$$-1 + 1 = 0$$

so they are perpendicular.

Its eigenvalues are never negative. Take a unit eigenvector $\mathbf{v}$ with eigenvalue λ. Then, one step per line:

$$\lambda = \lambda\, \mathbf{v}^{\mathsf{T}}\mathbf{v} = \mathbf{v}^{\mathsf{T}}(A^{\mathsf{T}}A\mathbf{v})$$

$$= (A\mathbf{v})^{\mathsf{T}}(A\mathbf{v})$$

$$= \|A\mathbf{v}\|^2 \geq 0$$

Here $\|A\mathbf{v}\|$ is the length of the output. Check with $\mathbf{v}_1$: the output $A\mathbf{v}_1$ is the vector $[3, 9]$ scaled by $1/\sqrt{2}$, and its squared length is the eigenvalue:

$$(9 + 81)/2 = 45$$

So the square root $\sigma = \sqrt{\lambda}$ is a real number of at least 0, and it is the length of $A\mathbf{v}$, as in Figure 2.

A symmetric matrix with no negative eigenvalues is **positive semi-definite** (G-1532). So $A^\mathsf{T}A$ is symmetric and positive semi-definite for every matrix A of any shape.

2.3 AA^T gives U, with the same numbers

> **Key point:** Multiplying in the other order, AA^T, cancels V instead and keeps U; its non-zero **eigenvalues** (G-665) are the same squared stretches.

In plain words. $A^\mathsf{T}A$ looks at the input side and finds the $\mathbf{v}$'s. Swapping the order looks at the output side and finds the $\mathbf{u}$'s.

On numbers. For A:

$$AA^\mathsf{T}$$

$$= \begin{bmatrix} 3 & 0 \\ 4 & 5 \end{bmatrix} \begin{bmatrix} 3 & 4 \\ 0 & 5 \end{bmatrix}$$

$$= \begin{bmatrix} 9 & 12 \\ 12 & 41 \end{bmatrix}$$

a different matrix from $A^\mathsf{T}A$. Its eigenvalues come from the determinant, as in finding eigenvalues, one step per line:

$$(9 - \lambda)(41 - \lambda) - 12 \times 12 = 0$$

$$\lambda^2 - 50\lambda + 369 - 144 = 0$$

$$\lambda^2 - 50\lambda + 225 = 0$$

$$(\lambda - 45)(\lambda - 5) = 0$$

So they are 45 and 5 again. The 50 is the diagonal sum:

$$9 + 41 = 50$$

$$45 + 5 = 50$$

and the 225 is the determinant:

$$9 \times 41 - 12 \times 12$$

$$= 369 - 144 = 225$$

$$45 \times 5 = 225$$

This holds for every square matrix: its eigenvalues add up to the diagonal sum and multiply to the determinant (Strang §6.1). Section 5 uses it as a shortcut.

The algebra, by the same steps as Section 2.1, using $V^\mathsf{T}V = I$:

$$AA^\mathsf{T} = U\Sigma V^\mathsf{T}\, V\Sigma^\mathsf{T}U^\mathsf{T}$$

$$= U\,\Sigma\Sigma^\mathsf{T}\,U^\mathsf{T}$$

So the left singular vectors $\mathbf{u}_i$ are the **eigenvectors** (G-666) of AA^T, and its non-zero eigenvalues are the same σ_i^2. The match of 45 and 5 is no accident: the middle factors $\Sigma\Sigma^\mathsf{T}$ (here) and $\Sigma^\mathsf{T}\Sigma$ (Section 2.1) are diagonal matrices with the same non-zero entries σ_i^2, so AA^T and $A^\mathsf{T}A$ share their non-zero eigenvalues for every matrix A.

3. The recipe, worked on a 2×2 matrix

> **Key point:** Find the eigenvectors and **eigenvalues** (G-665) of $A^\mathsf{T}A$, take square roots, then get each $\mathbf{u}_i$ by applying A to $\mathbf{v}_i$ and shrinking the result to length 1.

Figure 1 shows the four steps. We follow them for A with rows $[3, 0]$ and $[4, 5]$.

Step 1: form $A^\mathsf{T}A$. As in Section 2.1:

$$A^\mathsf{T}A = \begin{bmatrix} 25 & 20 \\ 20 & 25 \end{bmatrix}$$

Step 2: eigenvalues and eigenvectors of $A^\mathsf{T}A$.

In words: an eigenvalue λ is a number for which $A^\mathsf{T}A - \lambda I$ (subtract λ from the diagonal) squashes some arrow to zero; such a matrix has determinant 0.

On numbers:

$$\det \begin{bmatrix} 25-\lambda & 20 \\ 20 & 25-\lambda \end{bmatrix}$$

$$= (25-\lambda)^2 - 20 \times 20$$

$$(25-\lambda)^2 - 400 = 0$$

$$25 - \lambda = 20$$

or

$$25 - \lambda = -20$$

$$\lambda_1 = 45$$

$$\lambda_2 = 5$$

The eigenvalues are listed largest first.

$$\sigma_1 = \sqrt{45} = 3\sqrt{5} \approx 6.708$$

$$\sigma_2 = \sqrt{5} \approx 2.236$$

For each eigenvalue, find the line of arrows it squashes to zero:

λ	$A^\mathsf{T}A - \lambda I$	sends $[x, y]$ to zero when
45	rows $[-20, 20]$, $[20, -20]$	$x = y$
5	rows $[20, 20]$, $[20, 20]$	$y = -x$

Scaled to length 1 (divide by the length of $[1, 1]$):

$$\sqrt{1^2 + 1^2} = \sqrt{2}$$

$$\mathbf{v}_1$$

$$= \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\mathbf{v}_2$$

$$= \frac{1}{\sqrt{2}} \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

The two vectors are perpendicular, as Section 2.2 promised.

The formula:

$$\det(A^\mathsf{T} A - \lambda I) = 0$$

$$\sigma_i = \sqrt{\lambda_i}$$

Step 3: get each $\mathbf{u}_i$ from its $\mathbf{v}_i$.

In words: apply A to each right singular vector; the output points along an axis of the ellipse and has length σ_i. Shrink it back to length 1.

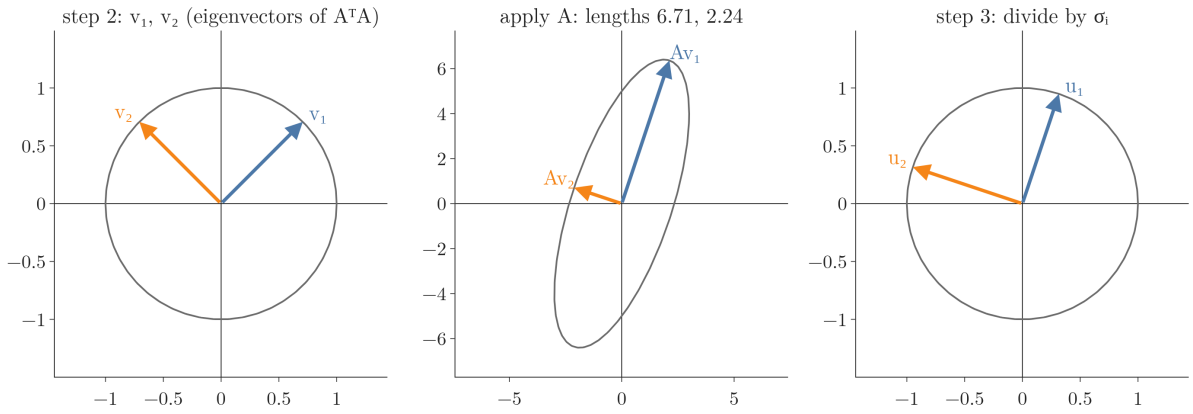


Figure 3: Steps 2 and 3 for A: the eigenvectors $\mathbf{v}_1, \mathbf{v}_2$ of $A^\mathsf{T} A$ (left), their images under A with lengths σ_1, σ_2 (middle), and the same images shrunk back to length 1, which are $\mathbf{u}_1, \mathbf{u}_2$ (right)

In Figure 3, watch the middle panel: $A\mathbf{v}_1$ and $A\mathbf{v}_2$ already point along the ellipse's axes, so dividing by their lengths only shrinks them back onto the circle (right).

On numbers:

$$A\mathbf{v}_1$$

$$= \frac{1}{\sqrt{2}} \begin{bmatrix} 3 \times 1 + 0 \times 1 \\ 4 \times 1 + 5 \times 1 \end{bmatrix}$$

$$= \frac{1}{\sqrt{2}} \begin{bmatrix} 3 \\ 9 \end{bmatrix}$$

$$\mathbf{u}_1$$

$$= \frac{1}{3\sqrt{5}} \cdot \frac{1}{\sqrt{2}} \begin{bmatrix} 3 \\ 9 \end{bmatrix}$$

$$= \frac{1}{\sqrt{10}} \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

$$A\mathbf{v}_2$$

$$= \frac{1}{\sqrt{2}} \begin{bmatrix} 3 \times (-1) + 0 \times 1 \\ 4 \times (-1) + 5 \times 1 \end{bmatrix}$$

$$= \frac{1}{\sqrt{2}} \begin{bmatrix} -3 \\ 1 \end{bmatrix}$$

$$\mathbf{u}_2$$

$$= \frac{1}{\sqrt{5}} \cdot \frac{1}{\sqrt{2}} \begin{bmatrix} -3 \\ 1 \end{bmatrix}$$

$$= \frac{1}{\sqrt{10}} \begin{bmatrix} -3 \\ 1 \end{bmatrix}$$

The formula: the step is the **singular value equation** (G-1814) $A\mathbf{v}_i = \sigma_i \mathbf{u}_i$ solved for $\mathbf{u}_i$:

$$\mathbf{u}_i = \frac{1}{\sigma_i} A\mathbf{v}_i$$

The **u**'s come out perpendicular without any extra work. Check on numbers: the dot product of $[1, 3]$ and $[-3, 1]$ is

$$-3 + 3 = 0$$

In symbols, one step per line:

$$(A\mathbf{v}_1)^\mathsf{T}(A\mathbf{v}_2) = \mathbf{v}_1^\mathsf{T}(A^\mathsf{T}A\mathbf{v}_2)$$

$$= \mathbf{v}_1^\mathsf{T}(\lambda_2\mathbf{v}_2)$$

$$= \lambda_2\, \mathbf{v}_1^\mathsf{T}\mathbf{v}_2$$

$$= 0$$

Step 4: complete U. Here there are already two **u**'s for a 2×2 matrix, so nothing is missing. Sections 5 and 7 need this step.

Check. Multiplying the factors back together, one product at a time:

$$U\Sigma$$

$$= \frac{1}{\sqrt{10}} \begin{bmatrix} 1 & -3 \\ 3 & 1 \end{bmatrix}$$

$$\cdot \begin{bmatrix} 3\sqrt{5} & 0 \\ 0 & \sqrt{5} \end{bmatrix}$$

$$= \frac{1}{\sqrt{2}} \begin{bmatrix} 3 & -3 \\ 9 & 1 \end{bmatrix}$$

$$U\Sigma V^\mathsf{T}$$

$$= \frac{1}{\sqrt{2}} \begin{bmatrix} 3 & -3 \\ 9 & 1 \end{bmatrix}$$

$$\cdot \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$$

$$= \frac{1}{2} \begin{bmatrix} 6 & 0 \\ 8 & 10 \end{bmatrix}$$

$$= \begin{bmatrix} 3 & 0 \\ 4 & 5 \end{bmatrix}$$

$$= A$$

Python: Checking each step with NumPy.

```
import numpy as np

A = np.array([[3, 0], [4, 5]])
lam, V = np.linalg.eigh(A.T @ A)    # eigh: for symmetric
lam, V = lam[::-1], V[:, ::-1]      # largest first
sigma = np.sqrt(lam)                # [6.708, 2.236]
U = A @ V / sigma                   # column i = A v_i / sigma_i
U @ np.diag(sigma) @ V.T            # gives back A
```

`A @ V / sigma` divides column i of AV by σ_i: NumPy matches the 1D array `sigma` with the columns.

4. The sign trap

Key point: Finding U and V separately as **eigenvectors** (G-666) can pick mismatched signs and give a wrong product; computing $\mathbf{u}_i = A\mathbf{v}_i/\sigma_i$ always matches them.

Section 2.3 suggests a shortcut: take the $\mathbf{u}$'s as the eigenvectors of AA^T, found on their own. For A, AA^T has rows $[9, 12]$ and $[12, 41]$, with **eigenvalues** (G-665) 45 and 5. Its eigenvectors are the lines through $[1, 3]$ and through $[3, -1]$.

Every non-zero multiple of an eigenvector is an eigenvector too, so $\frac{1}{\sqrt{10}}[3, -1]$ is as correct an eigenvector as $\frac{1}{\sqrt{10}}[-3, 1]$. But it is the negative of the $\mathbf{u}_2$ that belongs with our $\mathbf{v}_2$. Using it gives

$$\frac{1}{\sqrt{10}} \begin{bmatrix} 1 & 3 \\ 3 & -1 \end{bmatrix}$$

$$\cdot \begin{bmatrix} 3\sqrt{5} & 0 \\ 0 & \sqrt{5} \end{bmatrix}$$

$$\cdot \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 3 \\ 5 & 4 \end{bmatrix}$$

$$\neq A$$

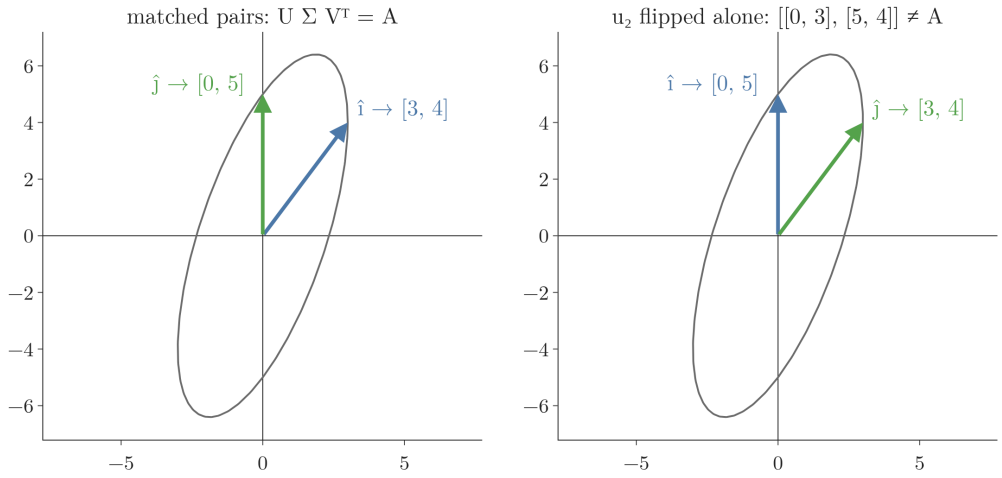


Figure 4: Left: the matched factors rebuild A. Right: the same factors with only $\mathbf{u}_2$ flipped. The ellipse is identical, but $\hat{\imath}$ and $\hat{\jmath}$ (the unit arrows along x and y, $[1, 0]$ and $[0, 1]$) land in swapped places

Each piece is a valid eigenvector, yet the product is a different matrix. Figure 4 shows why the mistake is easy to miss: both products have the singular values 6.71 and 2.24, so they draw the same ellipse; only where each input lands has changed. The pairs $(\mathbf{u}_i, \mathbf{v}_i)$ must match: flipping both signs is allowed, flipping one is not (see ordering and sign). The eigenvalue problem for AA^T knows nothing about V, so it cannot keep the pairs matched.

The fix is Step 3: compute $\mathbf{u}_i = A\mathbf{v}_i/\sigma_i$ from the $\mathbf{v}_i$ already chosen. Then $A\mathbf{v}_i = \sigma_i\mathbf{u}_i$ holds by construction, whatever sign each $\mathbf{v}_i$ was given.

5. A matrix of rank 1

> **Key point:** A rank-1 matrix has one non-zero singular value; the zero one has no $\mathbf{u}$ from $A\mathbf{v}/\sigma$, so we complete U with a perpendicular unit vector.

Some matrices squash the plane flat. Take

$$C = \begin{bmatrix} 2 & 1 \\ 4 & 2 \end{bmatrix}$$

Its columns $[2, 4]$ and $[1, 2]$ both lie on the line through $[1, 2]$. Every output $C\mathbf{x}$ is a mix of the two columns, so every output lands on that one line: C squishes the whole plane onto it.

- The set of all outputs is the **column space** (G-414) of C: here, the line through $[1, 2]$.
- The number of dimensions of the column space is the **rank** (G-1627) of the matrix: here 1. (The second row is twice the first, the algebraic sign of the same fact.)
- While the plane squashes onto a line, a whole other line of inputs, the one through $[-1, 2]$, lands on the origin. The set of inputs that land on $\mathbf{0}$ is the **null space** (G-1362).

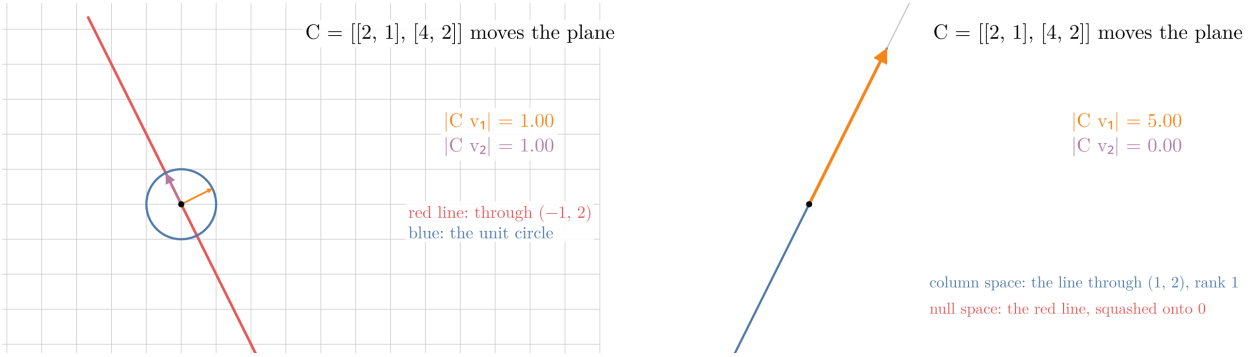


Figure 5: The rank-1 matrix C moves the grid: the whole plane, and the unit circle (blue) with it, squashes onto the line through $[1, 2]$, while every point of the red line through $[-1, 2]$ lands on the origin. $\mathbf{v}_1$ (orange) ends at length 5 and $\mathbf{v}_2$ (purple) at length 0. Idea after 3Blue1Brown, "Inverse matrices, column space and null space | Chapter 7, Essence of linear algebra"

In Figure 5, watch the red line while the grid moves: it shrinks towards the origin and vanishes, while the circle flattens into a segment. The two readouts end at 5 and 0, the two singular values the recipe now computes.

Step 1.

$$C^{\mathsf{T}}C$$

$$= \begin{bmatrix} 2 & 4 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 4 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 20 & 10 \\ 10 & 5 \end{bmatrix}$$

Step 2. The determinant and the diagonal sum (Section 2.3) give the two **eigenvalues** (G-665):

$$\det = 20 \times 5 - 10 \times 10 = 0$$

$$\implies \lambda_2 = 0$$

$$\lambda_1 + \lambda_2 = 20 + 5 = 25$$

$$\implies \lambda_1 = 25$$

$$\sigma_1 = \sqrt{25} = 5$$

$$\sigma_2 = 0$$

The **eigenvectors** (G-666) are

$$\mathbf{v}_1$$

$$= \frac{1}{\sqrt{5}} \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$\mathbf{v}_2$$

$$= \frac{1}{\sqrt{5}} \begin{bmatrix} -1 \\ 2 \end{bmatrix}$$

$\mathbf{v}_2$ is the direction C squishes to nothing:

$$C \begin{bmatrix} -1 \\ 2 \end{bmatrix}$$

$$= \begin{bmatrix} 2 \times (-1) + 1 \times 2 \\ 4 \times (-1) + 2 \times 2 \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

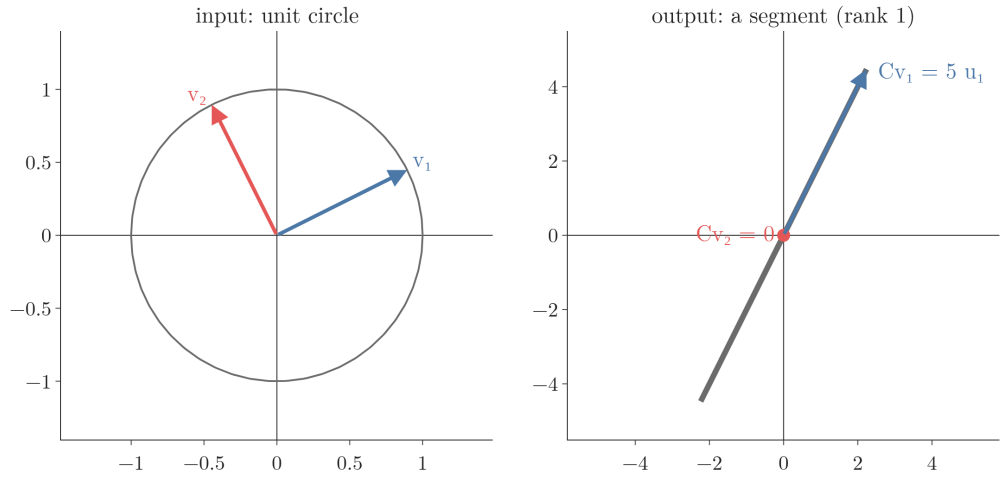


Figure 6: C flattens the unit circle onto a segment of the line through $[1, 2]$: $\mathbf{v}_1$ lands at length 5, $\mathbf{v}_2$ lands on the origin

In Figure 6, the ellipse of the 2×2 case has collapsed: its short axis σ_2 is 0, so the whole circle lands on one line, reaching 5 on each side of the origin.

Step 3. Only σ_1 is non-zero:

$$\mathbf{u}_1 = \frac{1}{5} C\mathbf{v}_1$$

$$= \frac{1}{5\sqrt{5}} \begin{bmatrix} 5 \\ 10 \end{bmatrix}$$

$$= \frac{1}{\sqrt{5}} \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

Step 4. Computing $\mathbf{u}_2$ as $C\mathbf{v}_2/\sigma_2$ would divide zero by zero. Any unit vector perpendicular to $\mathbf{u}_1$ completes U to an **orthogonal matrix** (G-1407); we take:

$$\mathbf{u}_2 = \frac{1}{\sqrt{5}}[-2, 1]$$

This $\mathbf{u}_2$ is multiplied by $\sigma_2 = 0$, so it never affects the product.

Result.

$$C$$

$$= \frac{1}{\sqrt{5}} \begin{bmatrix} 1 & -2 \\ 2 & 1 \end{bmatrix}$$

$$\cdot \begin{bmatrix} 5 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\cdot \frac{1}{\sqrt{5}} \begin{bmatrix} 2 & 1 \\ -1 & 2 \end{bmatrix}$$

Only the first column of U and the first row of V^{T} meet a non-zero number, so the product shrinks to

$$C = \sigma_1 \, \mathbf{u}_1 \, \mathbf{v}_1^{\mathsf{T}}$$

$$= 5 \cdot \frac{1}{\sqrt{5}} \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$\cdot \frac{1}{\sqrt{5}} \begin{bmatrix} 2 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \\ 2 \end{bmatrix} \begin{bmatrix} 2 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & 1 \\ 4 & 2 \end{bmatrix}$$

A column times a row is a whole matrix of rank 1. Every matrix can be built out of such pieces (see a matrix as a sum of rank-1 layers).

> **Python:** NumPy reports the zero singular value.
> `C = np.array([[2, 1], [4, 2]])`
> `np.linalg.svd(C)[1]` `# [5., 0.]`
> `np.linalg.matrix_rank(C)` `# 1`
> `matrix_rank` from **span** (G-1838) works exactly this way: it counts the singular values that are not (almost) zero.

6. The four subspaces

> **Key point:** Every matrix splits its inputs into two perpendicular parts, the directions it uses and the directions it squashes to zero, and splits its outputs into the directions it reaches and the directions it never reaches. The SVD gives perpendicular bases for all four.

The idea in plain words, on C. Look again at the rank-1 matrix C of Figures 5 and 6.

- On the input side, there are two perpendicular lines. Along the line through $[2, 1]$, C does all its work: it stretches $\mathbf{v}_1$ by 5. Along the line through $[-1, 2]$, C squashes everything to zero.
- On the output side, there are also two perpendicular lines. Every output lands on the line through $[1, 2]$. The perpendicular line through $[-2, 1]$ is never reached by any output.

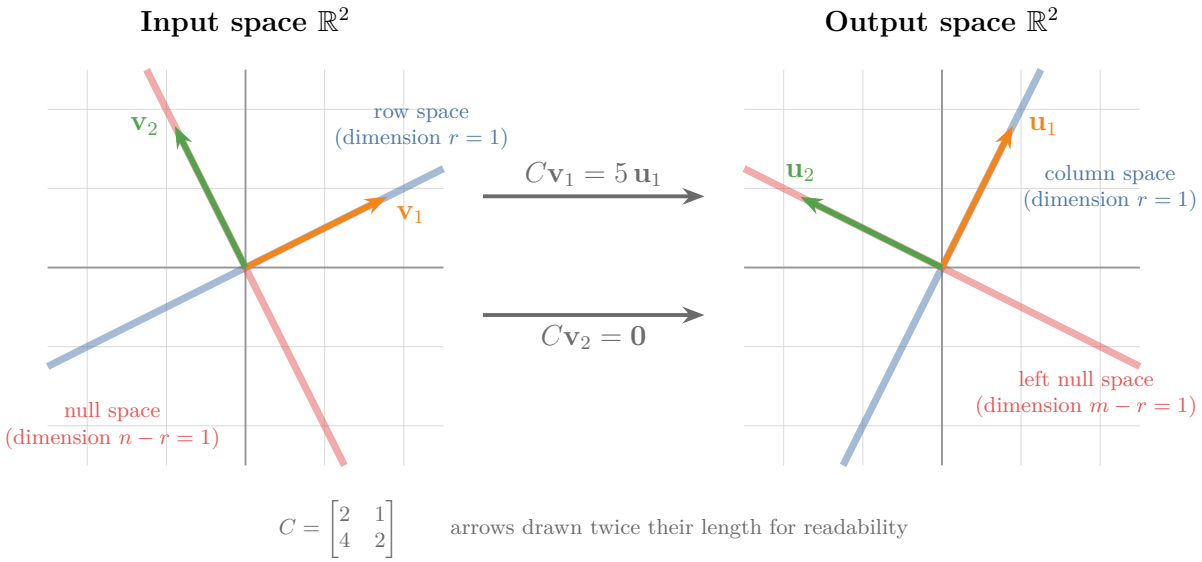


Figure 7: The SVD of C gives a basis for each of its four subspaces: input side (left) split into the row space and the null space, output side (right) split into the column space and the left null space

Figure 7 draws these four lines. On the left is the input plane, on the right the output plane; the titles write each plane as $\mathbb{R}^2$, the space of all pairs of numbers such as $(2, 1)$. The arrow $\mathbf{v}_1$ on the left is carried to the line of $\mathbf{u}_1$ on the right; the arrow $\mathbf{v}_2$ goes to the origin.

Worked check on numbers. One line per subspace:

$$C \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 5 \\ 10 \end{bmatrix}$$

$$= 5 \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

The used direction lands on the output line.

$$C \begin{bmatrix} -1 \\ 2 \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

The squashed direction goes to zero.

$$[1, 2] \cdot [-2, 1] = -2 + 2$$

$$= 0$$

The unreached direction is perpendicular to the output line.

The standard terms. Together these are the **four fundamental subspaces** (G-799) of a matrix (Strang §7.2). A **subspace** here is a line, plane or higher flat set through the origin. We write $\mathbb{R}^n$ for the space of all lists of n numbers: $\mathbb{R}^2$ holds pairs such as $(2, 1)$, $\mathbb{R}^3$ triples such as $(2, 1, 1)$. An $m \times n$ matrix (m rows, n columns) takes inputs from $\mathbb{R}^n$ and gives outputs in $\mathbb{R}^m$; for C, $m = n = 2$. For a matrix of **rank** (G-1627) r (there are r non-zero singular values):

Subspace	Where	Basis from the SVD	Dimension	For C
Column space (G-414) (span of the columns: every output)	output $\mathbb{R}^m$	$\mathbf{u}_1, \ldots, \mathbf{u}_r$	r	line through $[1, 2]$
Null space (G-1362) (vectors sent to **0**)	input $\mathbb{R}^n$	$\mathbf{v}_{r+1}, \ldots, \mathbf{v}_n$	$n - r$	line through $[-1, 2]$
Row space (G-1713) (span of the rows)	input $\mathbb{R}^n$	$\mathbf{v}_1, \ldots, \mathbf{v}_r$	r	line through $[2, 1]$

Subspace	Where	Basis from the SVD	Dimension	For C
Left null space (G-1078) (outputs never reached, perpendicular to them)	output $\mathbb{R}^m$	$\mathbf{u}_{r+1}, \ldots, \mathbf{u}_m$	$m - r$	line through $[-2, 1]$

Check: for C, $r = 1$, so each subspace has dimension 1 (a line), as the last column says.

The first two are the ones Figure 5 showed moving: the column space is where every output lands (the **span** (G-1838) of the columns, from reading a matrix as a picture), and the null space is everything the matrix squishes to the origin. The other two are their partners: the row space is the input directions perpendicular to the null space, and the left null space is the output directions perpendicular to the column space.

What makes the SVD's bases special is that they line up in pairs: A sends $\mathbf{v}_1$ to a multiple of $\mathbf{u}_1$, $\mathbf{v}_2$ to a multiple of $\mathbf{u}_2$, and so on, with no mixing. The null-space $\mathbf{v}$'s go to zero, which is where the zeros on the diagonal of Σ come from. Any method of making perpendicular bases could give bases of the four subspaces; only the SVD's bases also make the matrix diagonal.

7. A matrix that is not square

> **Key point:** A tall matrix turns short lists into long lists, so its outputs cannot fill the bigger space. $A^\mathsf{T}A$ is the small matrix to work with, and the missing columns of U point in the directions the outputs never reach.

The idea in plain words. Take the 3×2 matrix B of matrices that are not square:

$$B = \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}$$

It takes an input of 2 numbers and gives an output of 3 numbers. For example:

$$B \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 2+1 \\ 0+1 \\ 2+0 \end{bmatrix}$$

$$= \begin{bmatrix} 3 \\ 1 \\ 2 \end{bmatrix}$$

Every output is a mix of B's two columns, $[1, 0, 1]$ and $[1, 1, 0]$. Two arrows in 3D **span** (G-1838) a flat sheet through the origin, so all outputs lie on that sheet. One direction, sticking straight out of the sheet, is never reached.

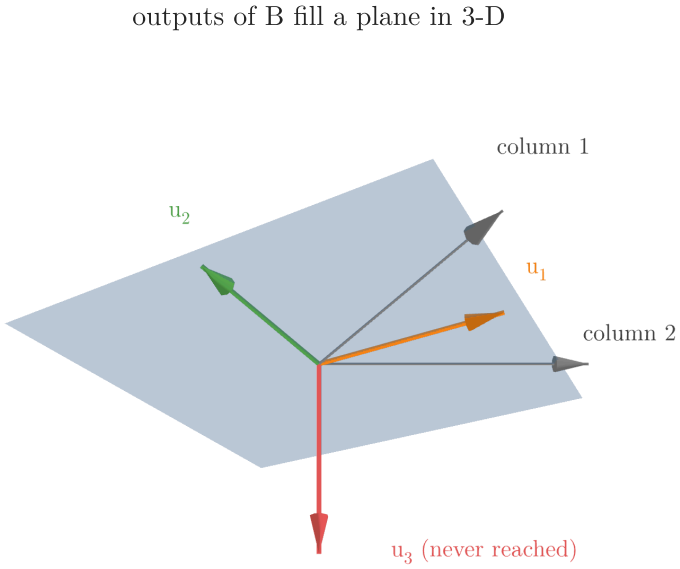


Figure 8: The outputs of B fill a plane in 3-D (blue): every output is a mix of B's two columns (grey). The left singular vectors u_1 (orange) and u_2 (green) lie in the plane; u_3 (red) sticks out of it at a right angle and is never reached

Figure 8 shows the sheet. The recipe will find $\mathbf{u}_1$ and $\mathbf{u}_2$ inside the sheet (Step 3), and Step 4 adds $\mathbf{u}_3$, the red arrow at a right angle to it.

Steps 1 and 2. $B^\mathsf{T}B$ is only 2×2:

$$B^\mathsf{T}B$$

$$= \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

$$\cdot \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

$$(2-\lambda)^2 - 1 = 0$$

$$2 - \lambda = 1$$

or

$$2 - \lambda = -1$$

$$\lambda_1 = 3$$

$$\lambda_2 = 1$$

So the singular values are:

$$\sigma_1 = \sqrt{3} \approx 1.732$$

$$\sigma_2 = 1$$

with these directions:

$$\mathbf{v}_1$$

$$= \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\mathbf{v}_2$$

$$= \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Step 3.

$$\mathbf{u}_1 = \frac{1}{\sqrt{3}} B \mathbf{v}_1$$

$$= \frac{1}{\sqrt{3}} \cdot \frac{1}{\sqrt{2}} \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}$$

$$= \frac{1}{\sqrt{6}} \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}$$

$$\approx \begin{bmatrix} 0.816 \\ 0.408 \\ 0.408 \end{bmatrix}$$

$$\mathbf{u}_2 = \frac{1}{1} B\mathbf{v}_2$$

$$= \frac{1}{\sqrt{2}} \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}$$

Step 4. U is 3×3 and needs a third column, perpendicular to both. Try $[1, -1, -1]$:

$$[1, -1, -1] \cdot [2, 1, 1] = 2 - 1 - 1 = 0$$

$$[1, -1, -1] \cdot [0, -1, 1] = 0 + 1 - 1 = 0$$

Both dot products are 0, so

$$\mathbf{u}_3 = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ -1 \\ -1 \end{bmatrix}$$

The vector $\mathbf{u}_3$ spans the **left null space** (G-1078), the one direction in 3D that B never reaches (the red arrow of Figure 8).

The other route, **eigenvectors** (G-666) of BB^T, would mean a 3×3 **eigenvalue** (G-665) problem (eigenvalues 3, 1 and 0), and would still risk the sign trap of Section 4. For a tall data matrix with thousands of rows, $A^\mathsf{T}A$ is the small one, so we always start from it.

8. How computers compute the SVD

> **Key point:** Forming $A^\mathsf{T}A$ squares the singular values, and small ones get lost in rounding. Libraries compute the SVD directly from A.

In plain words. A computer keeps only about 16 significant digits of each number. Squaring a tiny number makes it far tinier, and next to a big number it falls off the end of those 16 digits.

On numbers. Take a matrix like M in the box below: its big singular value is $\sigma_1 = 2$ and its small one is about 10^{-9}. Square both to get the **eigenvalues** (G-665) of $M^\mathsf{T}M$:

$$\sigma_1^2 = 2^2 = 4$$

$$\sigma_2^2 = (10^{-9})^2 = 10^{-18}$$

The small eigenvalue as a share of the big one:

$$10^{-18}/4$$

$$\approx 2.5 \times 10^{-19}$$

That is far below the 16th digit, so the small eigenvalue disappears in rounding.

> **Python:** A nearly rank-1 matrix: its second singular value is tiny but not zero.

```
M = np.array([[1, 1], [1, 1 + 1e-9]])
np.linalg.svd(M)[1]              # [2.0, 5.0e-10]: correct
np.sqrt(np.linalg.eigvalsh(M.T @ M))
# [0.0, 2.0]: the small one is lost
```

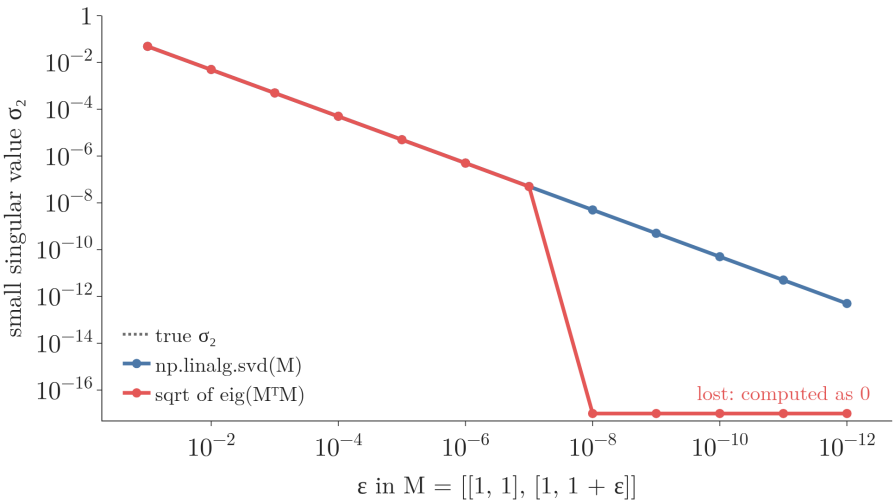


Figure 9: The small singular value of M as ε shrinks: `np.linalg.svd` (blue) follows the true value (dotted, under the blue line); the square root of the eigenvalue of $M^\mathsf{T}M$ (red) drops to 0 from $\varepsilon = 10^{-8}$ on

Figure 9 repeats the box for twelve values of the small number ε (the 10^{-9} in the code). Watch the red line: it agrees with the blue one while σ_2^2 still fits in the 16 digits next to $\sigma_1^2 = 4$, then falls off a cliff. So `np.linalg.svd` never forms $A^\mathsf{T}A$; NumPy calls the LAPACK routine `gesdd` (NumPy docs, `numpy.linalg.svd`). That routine reduces A itself step by step with orthogonal matrices, which do not magnify rounding errors, until the singular values can be read off (Trefethen and Bau, Lecture 31).

Extra: The ratio of the largest to the smallest singular value is the **condition number** (G-441) of a matrix:

$$\text{condition number} = \frac{\sigma_1}{\sigma_n}$$

For A:

$$\frac{6.71}{2.24} = 3$$

It measures how much errors in the input can be magnified by solving with that matrix. Forming $A^\mathsf{T}A$ squares it: a condition number of 10^8 becomes 10^{16}, which uses up all 16 digits (Trefethen and Bau, Lectures 12 and 19). The squaring is why the least-squares solvers of solving without an explicit inverse prefer to work from X itself rather than from $X^\mathsf{T}X$. `np.linalg.cond(A)` computes it.

9. Summary

Step	What we compute	For A (rows $[3,0]$, $[4,5]$)	For C (rows $[2,1]$, $[4,2]$)	Why
1	$A^\mathsf{T}A$	rows $[25,20]$, $[20,25]$	rows $[20,10]$, $[10,5]$	the turn U cancels, leaving a symmetric matrix we can solve by hand
2	eigenvalues, then $\sigma_i = \sqrt{\lambda_i}$	$45, 5$ give $6.708, 2.236$	$25, 0$ give $5, 0$	its eigenvalues are the squared stretches
2	eigenvectors $\mathbf{v}_i$	$\frac{1}{\sqrt{2}}[1,1]$, $\frac{1}{\sqrt{2}}[-1,1]$	$\frac{1}{\sqrt{5}}[2,1]$, $\frac{1}{\sqrt{5}}[-1,2]$	they are the directions of most and least stretch
3	$\mathbf{u}_i = A\mathbf{v}_i/\sigma_i$	$\frac{1}{\sqrt{10}}[1,3]$, $\frac{1}{\sqrt{10}}[-3,1]$	$\frac{1}{\sqrt{5}}[1,2]$	keeps each $\mathbf{u}_i$ matched in sign to its $\mathbf{v}_i$
4	complete U	nothing missing	$\frac{1}{\sqrt{5}}[-2,1]$	a zero σ_i gives no $\mathbf{u}_i$ to divide out

- $A^\mathsf{T}A = V\Sigma^\mathsf{T}\Sigma V^\mathsf{T}$: **eigenvectors** (G-666) are the $\mathbf{v}_i$, **eigenvalues** (G-665) the σ_i^2, because $U^\mathsf{T}U = I$ makes the turn U cancel.
- $A^\mathsf{T}A$ is always symmetric and **positive semi-definite** (G-1532), so this always works.
- Get the $\mathbf{u}_i$ from $A\mathbf{v}_i/\sigma_i$, not from a separate eigen-problem: that keeps the signs matched.
- Zero singular values belong to the **null space** (G-1362); the missing $\mathbf{u}$'s are any perpendicular completion, because $A\mathbf{v}_i/\sigma_i$ cannot be computed when $\sigma_i = 0$.
- The SVD gives perpendicular bases for the row space, null space, column space and **left null space** (G-1078), and the bases line up in pairs, so the matrix is diagonal in them.
- Computers compute the SVD from A directly, because $A^\mathsf{T}A$ loses small singular values.
- So the SVD can be found by hand from the eigenvectors and eigenvalues of $A^\mathsf{T}A$, the matrix that keeps the stretching and forgets the final turn.

10. Sources

Built from

- 3Blue1Brown (Sanderson, G.), "Inverse matrices, column space and null space | Chapter 7, Essence of linear algebra", YouTube, https://www.youtube.com/watch?v=uQhTuRlWMxw
- Khan Academy (Khan, S.), "Column space of a matrix", YouTube, https://www.youtube.com/watch?v=
- Khan Academy (Khan, S.), "Rowspace and left nullspace", YouTube, https://www.youtube.com/watch

- Strang, G., MIT OpenCourseWare 18.06 *Linear Algebra*, "Lecture 29: Singular value decomposition" (video, recorded 1999), ocw.mit.edu/courses/18-06-linear-algebra-spring-2010/resources/lecture-29-singular-value-decomposition. The recipe, the sign trap, the rank-1 example and the four subspaces (Sections 3 to 6).

Other references

- Strang, G. (2016). *Introduction to Linear Algebra*, 5th ed. Wellesley-Cambridge Press. Section 2.7 (transposes: the transpose of a product); Section 6.1 (eigenvalues add up to the diagonal sum and multiply to the determinant); Section 7.2, bases and matrices in the SVD: the recipe and the non-square example, which no video in the list works through.
- Deisenroth, M. P., Faisal, A. A. and Ong, C. S. (2020). *Mathematics for Machine Learning*. Cambridge University Press. Section 4.5.2 (MML).
- NumPy documentation. `numpy.linalg.svd` (uses LAPACK `gesdd`).
- Trefethen, L. N. and Bau, D. (1997). *Numerical Linear Algebra*. SIAM. Lecture 12 (conditioning), Lecture 19 (least squares and the normal equations), Lecture 31 (computing the SVD).

11. Key terms

Terms taught in this Note come first; linked terms are recaps, taught in the Note the link opens.

Term	Meaning
Rank (of a matrix) (G-1627)	The number of linearly independent columns of a matrix, that is, how many directions its columns really span; a column that is a combination of others does not add to it.
Null space (G-1362)	All vectors a matrix sends to $\mathbf{0}$; spanned by the $\mathbf{v}_i$ with $\sigma_i = 0$.
Four fundamental subspaces (G-799)	Four sets of directions tied to every matrix that show what it does: which input directions it uses, which it squashes to zero, which outputs it can reach and which it never reaches. They are the row space, null space, column space and left null space.
Row space (G-1713)	All the combinations of a matrix's rows (their span), a subspace of the input space; the SVD's right singular vectors $\mathbf{v}_i$ with $\sigma_i > 0$ are a perpendicular basis of it.

Term	Meaning
Left null space (G-1078)	The set of output vectors perpendicular to every column of a matrix, so the matrix can never produce them; in the SVD it is spanned by the left singular vectors $\mathbf{u}_{r+1}, \ldots, \mathbf{u}_m$.
Condition number (G-441)	A number that says how much a matrix can magnify small errors, such as rounding errors, when we solve with it; a large value means the answer is sensitive. It is the largest singular value over the smallest, σ_1/σ_n.
$A^\mathsf{T}A$ (G-22)	The matrix A transposed times A, used to find the SVD by hand: it is symmetric, its eigenvectors are the right singular vectors of A and its eigenvalues are the squared singular values.
Eigenvalue (G-665)	The number by which a matrix stretches or shrinks one of its eigenvectors: $A\mathbf{v} = \lambda\mathbf{v}$. In PCA, the eigenvalues of the covariance matrix are the variances along the principal components.
Transpose (G-2012)	A matrix or vector with rows and columns swapped; it turns a column vector u into a row u^T, so $u^\mathsf{T}x$ is the dot product.
Right singular vector ($\mathbf{v}_i$) (G-1692)	A column $\mathbf{v}_i$ of V in the SVD: a unit-length input direction that the matrix sends to σ_i times the matching left singular vector $\mathbf{u}_i$.
Diagonal matrix (G-601)	A matrix with zeros everywhere off the diagonal; it scales each axis by its own factor.
Singular value (σ_i) (G-1812)	A stretch factor of a matrix: the length of $A\mathbf{v}_i$; never negative, listed largest first.
Left singular vector ($\mathbf{u}_i$) (G-1079)	One of the output directions of the SVD: the unit vector that a right singular vector lands on, stretched by its singular value ($A\mathbf{v}_i = \sigma_i\mathbf{u}_i$); the columns of U.
Symmetric matrix (G-1932)	A square matrix equal to its own transpose, such as $X^\mathsf{T}X$ or a covariance matrix; it always has a full set of perpendicular eigenvectors.
Eigenvector (G-666)	A non-zero vector that a matrix only stretches or shrinks, without turning it off its own line: $A\mathbf{v} = \lambda\mathbf{v}$. In PCA, the eigenvectors of the covariance matrix are the directions of the principal components.

Term	Meaning
Identity matrix (G-915)	The matrix that leaves every vector unchanged.
Orthogonal matrix (G-1407)	A square matrix that only rotates or flips: its columns have length 1 and are all perpendicular (orthonormal), and its inverse is its transpose.
Positive semi-definite (G-1532)	A symmetric matrix that never curves downward in any direction: its eigenvalues are all 0 or positive. For such a matrix the eigenvectors are orthonormal, so its eigen-decomposition and its SVD are the same.
Singular value equation (G-1814)	The equation $A\mathbf{v}_i = \sigma_i \mathbf{u}_i$: the matrix sends each input direction $\mathbf{v}_i$ to its output direction $\mathbf{u}_i$, stretched by the singular value σ_i; it is used to compute each $\mathbf{u}_i$.
Column space (G-414)	Every output a matrix can produce: all the vectors made by scaling and adding its columns (the span of its columns).
Span (G-1838)	The set of all linear combinations of some vectors: every point reachable by scaling and adding them, such as the whole plane for $\hat{\imath}$ and $\hat{\jmath}$.